The Dry Ten Martini Problem at Criticality

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In memory of Matthew S. Brennan, 1994–2021

Abstract

We solve the dry ten martini problem (*Lean-verified* this time) by showing that every labeled almost Mathieu gap is open for $\lambda \neq 0$. In particular, we prove that the maximum logarithmic height of each labeled gap is non-decreasing in the coupling λ and extend an initial open *seed gap* for each gap label n from $\lambda \in (0, \lambda_n)$ to all $\lambda > 0$. The non-decreasing property follows from restoring time-reversal symmetry to the almost Mathieu operator H by pairing it with its magnetic inverse $\bar{H}$. This allows us to express the logarithmic coupling derivative of the maximal gap height as the expectation value of a non-negative operator, yielding the required monotonicity.

Lean labels. L^*: same proof; LE: cited inputs replaced by checked counterparts; W: unverified, unused background. Dependencies retain their own status. Appendix A gives the proof correspondence.

1 Introduction

We consider the almost Mathieu operator (AMO) $h_{\lambda,\alpha,\varphi} \in \mathcal{B}(\ell^2(\mathbb{Z}))$, with coupling $\lambda \in \mathbb{R} \setminus \{0\}$, irrational frequency α, and phase $\varphi \in \mathbb{R}/(2\pi\mathbb{Z})$, given by

$$(h_{\lambda,\alpha,\varphi}\psi)_j = \psi_{j+1} + \psi_{j-1} + 2\lambda\cos(2\pi\alpha j + \varphi)\psi_j. \tag{1.1}$$

Its spectrum $\Sigma_{\lambda,\alpha}$ is phase independent [6]. The bounded components of its complement are the *open gaps*. From now on, α is fixed and $\Sigma_\lambda = \Sigma_{\lambda,\alpha}$. Write $h_{\lambda,\varphi} = h_{\lambda,\alpha,\varphi}$, and let $(e_j)_{j\in\mathbb{Z}}$ be the canonical basis of $\ell^2(\mathbb{Z})$.

Harper introduced the model for conduction electrons in a magnetic field [21]; it later became a lattice model of the integer quantum Hall effect [39]. Spinless electrons hop between neighboring sites of $\mathbb{Z}^2$ with flux $\Theta = 2\pi\alpha$ through each plaquette. On $\ell^2(\mathbb{Z}^2)$, with canonical basis $\tilde{e}_{n,m}$, the Landau-gauge magnetic translations are

$$U\tilde{e}_{n,m} = \tilde{e}_{n-1,m}, \qquad V\tilde{e}_{n,m} = e^{\mathrm{i}\Theta n}\tilde{e}_{n,m-1}. \tag{1.2}$$

They satisfy $UV = e^{\mathrm{i}\Theta}VU$. We define the *magnetic rotation algebra* to be the unital C^*-algebra generated by U, V [35, p. 417]:

$$\mathcal{A}_\alpha := C^*(U,V) = \overline{\mathrm{span}\{U^pV^q : p,q \in \mathbb{Z}\}}^{\|\cdot\|} \subset \mathcal{B}(\ell^2(\mathbb{Z}^2)). \tag{1.3}$$

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Write $e_{0,0} := \tilde{e}_{0,0}$ for the origin. The Harper operator and its resolvent are

$$H_\lambda = U + U^* + \lambda(V + V^*), \qquad R_\lambda(z) := (z - H_\lambda)^{-1} \quad (z \notin \Sigma_\lambda). \tag{1.4}$$

Partial Fourier transformation in the second coordinate is the unitary map $(\mathcal{F}_y \tilde{e}_{n,m})(\varphi) = e^{-\mathrm{i}m\varphi} e_n$, with measure $\mathrm{d}\varphi/(2\pi)$, decomposing H_λ into the fibers (1.1), so $\sigma(H_\lambda) = \Sigma_\lambda$. For $A \in \mathcal{A}_\alpha$ with fibers $A(\varphi)$, the *canonical trace* is

$$\tau_\alpha(A) := \langle e_{0,0}, A e_{0,0} \rangle = \int_0^{2\pi} \langle e_0, A(\varphi) e_0 \rangle \frac{\mathrm{d}\varphi}{2\pi}. \tag{1.5}$$

It is positive and normalized, with $\tau_\alpha(AB) = \tau_\alpha(BA)$ for $A, B \in \mathcal{A}_\alpha$ [9, Secs. 3.1 and 3.6]. We use (1.5) to write the fiber integrals in trace notation. Fourier transforming the other coordinate, or making a magnetic quarter-turn in two dimensions, gives Aubry–André duality [1]:

$$\mathcal{R}\tilde{e}_{n,m} = e^{\mathrm{i}\Theta nm} \tilde{e}_{-m,n}, \qquad \mathcal{R} H_\lambda \mathcal{R}^* = \lambda H_{1/\lambda}. \tag{1.6}$$

The sign involution is

$$\mathcal{S}\tilde{e}_{n,m} = (-1)^m \tilde{e}_{n,m}, \qquad \mathcal{S} H_\lambda \mathcal{S}^* = H_{-\lambda}. \tag{1.7}$$

Both unitaries fix $e_{0,0}$. We write $k_y := \varphi$ for the Fourier variable dual to the second lattice coordinate, y, and k_x for the Fourier variable dual to the first, x.

1.1 Fermi projectors and gap labels

For Harper's noninteracting, particle-number-conserving model, an initial Slater-determinant state evolves by propagating its occupied single-particle orbitals under the action of the one-particle magnetic Hamiltonian [14, Sec. 1]. The zero-temperature single-particle occupied subspace is given by the Fermi projector:

Definition 1.1 (Fermi projector and density of states)**.** For real $E \notin \Sigma_\lambda$, choose a counterclockwise contour $\Gamma_{\lambda,E} \subset \mathbb{C} \setminus \Sigma_\lambda$ enclosing precisely the spectrum below E. The Fermi projector is given by the Riesz formula,

$$\chi_{H_\lambda}(E) := \frac{1}{2\pi\mathrm{i}} \oint_{\Gamma_{\lambda,E}} R_\lambda(z)\, \mathrm{d}z \in \mathcal{A}_\alpha. \tag{1.8}$$

Spectral calculus identifies this with the projection onto energies at most E. For any bounded self-adjoint A, the Borel formula

$$\chi_A(E) := \mathbf{1}_{(-\infty,E]}(A), \tag{1.9}$$

with $\mathbf{1}_S$ the indicator function on the set S, extends the definition to all real E. The origin pairings of these projectors define the integrated density of states and associated probability measure ν_λ:

$$N_\lambda(E) := \langle e_{0,0}, \chi_{H_\lambda}(E) e_{0,0} \rangle, \qquad \nu_\lambda((-\infty, E]) = N_\lambda(E). \tag{1.10}$$

Remark 1.2. The projector construction (1.8)–(1.10) and (1.5) give $N_\lambda(E) = \tau_\alpha(\chi_{H_\lambda}(E))$ in a gap, and spectral calculus gives $\int f \,\mathrm{d}\nu_\lambda = \tau_\alpha(f(H_\lambda))$ for continuous f. At spectral energies, (1.10) uses the origin pairing of the Borel projector, which need not belong to $\mathcal{A}_\alpha$.

Writing $\{x\} := x - \lfloor x \rfloor$ for the fractional part, we relate $N_\lambda(E)$ to α:

Theorem 1.3 (Gap labeling [7]; W)**.** *On each bounded gap, $N_\lambda(E) = \{n\alpha\}$ for a unique $n \in \mathbb{Z}\setminus\{0\}$.*

Proof sketch. The Riesz projector belongs to $\mathcal{A}_\alpha$ by (1.8), and its trace lies in $(\mathbb{Z} + \alpha\mathbb{Z}) \cap [0, 1]$ [30, 35]. The projector is constant on a gap; full support of ν_λ gives distinct IDS values on distinct gaps [6], and $\alpha \notin \mathbb{Q}$ makes the labels unique. □

With compatible orientation conventions, the integer n is the Chern number governing the quantized Hall conductance [39, 9]. Bulk–edge correspondence relates it to boundary spectral flow and quantized edge conductance [31, 25], modelling the integer quantum Hall effect.

1.2 The Ten Martini problems

In 1981, Kac famously asked whether "all gaps are there" and offered ten martinis for a solution [26, Remark 1.1(e)]. Simon published the challenge [36], leading to decades of work on AMO spectrum. The question came to be studied in two forms: the Ten Martini problem asks for Cantor spectrum, so that open gaps accumulate at every spectral energy; the *dry* Ten Martini problem retains the full gap-opening requirement, that is, every label in Theorem 1.3 must be attained by an open gap.

In a milestone result, Avila and Jitomirskaya solved the Ten Martini problem [3] and showed that open gaps accumulate at every spectral energy:

Theorem 1.4 (Ten Martini; Avila–Jitomirskaya; W)**.** *For every irrational frequency and every nonzero coupling, Σ_λ is a Cantor set.*

Reducibility, localization, and cocycle methods gave successive gap-opening results [15, 32, 4, 2]; Avila–You–Zhou proved the dry theorem for every noncritical AMO coupling [5]. Building the Type I framework [19], Ge–Jitomirskaya–You established that analytic Type I one-frequency quasiperiodic Schrödinger operators have Cantor spectrum for every irrational frequency, with stability in the appropriate analytic topology [20].

For the stronger dry statement, Li [27] and Ge [18] gave different proofs that, for every irrational frequency and real analytic potential, every subcritical Type I spectral energy satisfying the gap-labelling condition is an endpoint of an open gap. Ge's proof supplies the details of the argument announced in [20, Remark 1.3]. In the supercritical regime, Li–Xu–Zhou proved the corresponding Type I dry statement and, in particular, stability of all supercritical AMO gaps under sufficiently small analytic perturbations [28].

At critical coupling, Choi–Elliott–Yui proved full gap opening for a subclass of Liouville frequencies [12]. Figueras and Puig obtained rigorous computer-assisted proofs of individual labeled gaps at specific irrational frequencies, together with bounds on their widths [16]. However, the full dry Ten Martini problem for arbitrary irrational frequency at critical coupling remained open. Although this constitutes a single coupling value $\lambda = 1$, it marks a change of spectral type: for Diophantine frequencies and almost every phase, the spectrum is purely absolutely continuous for $0 < \lambda < 1$, pure point for $\lambda > 1$, and purely singular continuous at $\lambda = 1$ [23, 24]. Moreover, singular continuous spectra are Baire generic in suitable families of self-adjoint operators [37], suggesting the failure of existing methods at $\lambda = 1$ constituted a fundamental obstruction.

In this manuscript, motivated by the extensive literature on invertible phases of matter [40, 17], we pair the AMO with its time-reversed magnetic action to restore time-reversal symmetry on the spectral source space[1]. We use the resulting magnetic balance identity to arrive at the bound in Theorem 4.1, *closing the gap*[2] and proving that all gaps are open[3]:

[1] The same time-reversal symmetric pairing was used to study phase-averaged transport for the critical AMO [10] and in multiple works on quasiperiodic operators by Riedel [33, 34].

[2] Phrasing proposed by Wencai Liu.

[3] See Fig 1 for an annotated sketch of the full argument.

Theorem 1.5 (Dry Ten Martini; *LE*)**.** *For every irrational α, every $\lambda \neq 0$, and every $n \in \mathbb{Z} \setminus \{0\}$, there is a bounded open gap on which $N_\lambda(E) = \{n\alpha\}$.*

The noncritical case is due to Avila–You–Zhou [5]. However, we highlight the scope of the balance identity by treating all nonzero couplings directly. Indeed, the magnetic balance refines Aubry–André duality [1] into a conserved SU(2) doublet for the centered dual pair (Corollary 4.3).

2 Restoring Time-Reversal Symmetry

Reversing the magnetic phases conjugates the hopping amplitudes. We implement this in the symmetric gauge, starting from the Landau basis (1.2):

$$e_{n,m} := e^{-\mathrm{i}\pi\alpha nm}\tilde{e}_{n,m}, \qquad (\mathcal{F}_y e_{n,m})(k_y) = e^{-\mathrm{i}\pi\alpha nm}e^{-\mathrm{i}mk_y}e_n. \tag{2.1}$$

Define the *magnetic inversion* $\mathcal{C}$ to be coefficientwise complex conjugation in this basis, so $\mathcal{C}e_{n,m} = e_{n,m}$. In the Landau basis, for $v \in \ell^2(\mathbb{Z}^2)$, this reads

$$\langle \tilde{e}_{n,m}, \mathcal{C}v\rangle = e^{-\mathrm{i}\Theta nm}\overline{\langle \tilde{e}_{n,m}, v\rangle}. \tag{2.2}$$

Thus $\mathcal{C}$ is an anti-unitary involution, with $\mathcal{C}^2 = I$, $\mathcal{C}^* = \mathcal{C}$, and $\mathcal{C}e_{0,0} = e_{0,0}$.

Definition 2.1 (Magnetic inverse)**.** The *magnetic-inverse Hamiltonian* is

$$\bar{H}_\lambda := \mathcal{C}H_\lambda\mathcal{C}^* \in \mathcal{C}\mathcal{A}_\alpha\mathcal{C}^*. \tag{2.3}$$

The algebra $\mathcal{C}\mathcal{A}_\alpha\mathcal{C}^*$ has opposite flux, since

$$(\mathcal{C}U\mathcal{C}^*)(\mathcal{C}V\mathcal{C}^*) = e^{-\mathrm{i}\Theta}(\mathcal{C}V\mathcal{C}^*)(\mathcal{C}U\mathcal{C}^*). \tag{2.4}$$

For $t \in \mathbb{R}$, anti-unitarity reverses time and exchanges the Fermi projectors:

$$\mathcal{C}e^{-\mathrm{i}tH_\lambda}\mathcal{C}^* = e^{\mathrm{i}t\bar{H}_\lambda}, \qquad \mathcal{C}\chi_{H_\lambda}(E)\mathcal{C}^* = \chi_{\bar{H}_\lambda}(E). \tag{2.5}$$

Let $\mathcal{V}_\lambda$ be the closed span of $\chi_{H_\lambda}(E)e_{0,0}$ over real E,

Lemma 2.2 (Stationarity; L^*)**.** *The operators H_λ and $\bar{H}_\lambda$ leave $\mathcal{V}_\lambda$ invariant and coincide on it. In particular, for every $E \in \mathbb{R}$,*

$$(H_\lambda - \bar{H}_\lambda)\chi_{H_\lambda}(E)e_{0,0} = 0. \tag{2.6}$$

Proof. In the symmetric basis,

$$Ue_{n,m} = e^{-\mathrm{i}\pi\alpha m}e_{n-1,m}, \qquad Ve_{n,m} = e^{\mathrm{i}\pi\alpha n}e_{n,m-1}. \tag{2.7}$$

So, the reflection $e_{n,m} \mapsto e_{n,-m}$ sends U to $\mathcal{C}U\mathcal{C}^*$ and V to $\mathcal{C}V^*\mathcal{C}^*$, and hence sends H_λ to $\bar{H}_\lambda$.

Writing $[A, B] := AB - BA$, the original and reflected shifts satisfy $[U, \mathcal{C}U\mathcal{C}^*] = [V, \mathcal{C}V\mathcal{C}^*] = 0$, and by direct computation,

$$\begin{aligned} U(\mathcal{C}V\mathcal{C}^*)e_{n,m} &= (\mathcal{C}V\mathcal{C}^*)Ue_{n,m} = e^{-\mathrm{i}\pi\alpha(n+m-1)}e_{n-1,m-1}, \\ V(\mathcal{C}U\mathcal{C}^*)e_{n,m} &= (\mathcal{C}U\mathcal{C}^*)Ve_{n,m} = e^{\mathrm{i}\pi\alpha(n+m-1)}e_{n-1,m-1}. \end{aligned} \tag{2.8}$$

By unitarity, the same is true for adjoints, and at the origin, $\exp(\mathrm{i}\pi\alpha 0) = 1$. So,

$$[H_\lambda, \bar{H}_\lambda] = 0 \qquad \text{and} \qquad H_\lambda e_{0,0} = \bar{H}_\lambda e_{0,0}. \tag{2.9}$$

This implies commutation of $\bar{H}_\lambda$ with continuous functions of H_λ by polynomial approximation, and with spectral projections of H_λ by strong limits, i. e., $\chi_{H_\lambda}(E)$ commutes with $\bar{H}_\lambda$. Hence,

$$(H_\lambda - \bar{H}_\lambda)\chi_{H_\lambda}(E)e_{0,0} = \chi_{H_\lambda}(E)(H_\lambda - \bar{H}_\lambda)e_{0,0} = 0. \tag{2.10}$$

where the last equality follows from the second part of (2.9). The bounded operator $H_\lambda - \bar{H}_\lambda$ therefore vanishes on every generator of $\mathcal{V}_\lambda$, hence on their closed span. This space is invariant under H_λ by spectral calculus, and therefore also under $\bar{H}_\lambda$. □

Define $H_{\pm,\lambda} = \frac{1}{2}(H_\lambda \pm \bar{H}_\lambda)$. The symmetrized Hamiltonian is invariant under $\mathcal{C}$ by construction:

$$\mathcal{C}H_{+,\lambda}\mathcal{C}^* = H_{+,\lambda}. \tag{2.11}$$

Stationarity makes this symmetrization exact on the spectral source space:

$$H_{-,\lambda}|_{\mathcal{V}_\lambda} = 0, \qquad H_{+,\lambda}|_{\mathcal{V}_\lambda} = H_\lambda|_{\mathcal{V}_\lambda}, \qquad \sigma(H_{+,\lambda}|_{\mathcal{V}_\lambda}) = \Sigma_\lambda. \tag{2.12}$$

The spectral equality follows because the origin measure is ν_λ, whose support is Σ_λ [6]. This space is also $\mathcal{C}$-invariant. Indeed, $\mathcal{C}$ fixes $e_{0,0}$,

$$\mathcal{C}\chi_{H_\lambda}(E)e_{0,0} = \chi_{\bar{H}_\lambda}(E)e_{0,0} = \chi_{H_\lambda}(E)e_{0,0}. \tag{2.13}$$

Consequently, the dynamics generated by H_λ on $\mathcal{V}_\lambda$ satisfy

$$\mathcal{C}e^{-\mathrm{i}tH_\lambda}v = e^{\mathrm{i}tH_\lambda}\mathcal{C}v, \qquad v \in \mathcal{V}_\lambda, \quad t \in \mathbb{R}. \tag{2.14}$$

This is time-reversal symmetry on the spectral source space.

In the symmetric basis, the even and odd combinations under $\mathcal{C}$ act by

$$2H_{\pm,\lambda}e_{n,m} = \sum_{\rho=\pm 1}\Big[(e^{\mathrm{i}\rho\pi\alpha m} \pm e^{-\mathrm{i}\rho\pi\alpha m})e_{n+\rho,m} + \lambda(e^{-\mathrm{i}\rho\pi\alpha n} \pm e^{\mathrm{i}\rho\pi\alpha n})e_{n,m+\rho}\Big]. \tag{2.15}$$

The quarter-turn satisfies $\mathcal{R}H_{\pm,\lambda}\mathcal{R}^* = \lambda H_{\pm,1/\lambda}$. We introduce an observable that changes sign when the two hopping directions are exchanged:

Definition 2.3 (Sector parity)**.** Define

$$L_z e_{n,m} = \operatorname{sgn}(n^2 - m^2)e_{n,m}, \qquad \operatorname{sgn}(0) = 0. \tag{2.16}$$

Under Aubry duality, $\mathcal{R}L_z\mathcal{R}^* = -L_z$, and under time reversal, $\mathcal{C}L_z\mathcal{C}^* = L_z$.

The diagonals $m = \pm n$ divide the lattice into four sectors. We call the two regions with $|n| > |m|$ the horizontal sectors and those with $|m| > |n|$ the vertical sectors. Thus L_z equals $+1$ on the horizontal sectors, -1 on the vertical sectors, and 0 on the separating diagonals. Its commutator with the hopping operator is supported on bonds meeting those diagonals.

Stationarity cancels the nonzero diagonal contributions in the following commutator pairings, leaving only the origin.

Lemma 2.4 (Magnetic balance; L^*)**.** *For $v, w \in \ker H_{-,\lambda}$,*

$$\langle v, [L_z, H_{+,\lambda}]w\rangle = \langle v, L_z H_{+,\lambda} e_{0,0}\rangle\langle e_{0,0}, w\rangle - \langle v, e_{0,0}\rangle\langle L_z H_{+,\lambda} e_{0,0}, w\rangle. \tag{2.17}$$

Proof. Let Π_d project onto the separating diagonals $|n| = |m|$. Every bond that changes sector touches a diagonal, so

$$[L_z, H_{+,\lambda}] = L_z H_{+,\lambda}\Pi_d - \Pi_d H_{+,\lambda} L_z. \tag{2.18}$$

For $n \neq 0$, applying L_z to the diagonal hops in (2.15) and using (2.16) gives

$$\sin(\pi\alpha n) L_z H_{+,\lambda} e_{n,\pm n} = \mp \mathrm{i}\,\mathrm{sgn}(n)\cos(\pi\alpha n) H_{-,\lambda} e_{n,\pm n}. \tag{2.19}$$

Since $\sin(\pi\alpha n) \neq 0$ and $v, w \in \ker H_{-,\lambda}$, (2.19) gives

$$\langle v, L_z H_{+,\lambda} e_{n,\pm n}\rangle = \langle w, L_z H_{+,\lambda} e_{n,\pm n}\rangle = 0 \qquad (n \neq 0). \tag{2.20}$$

At the origin $H_{-,\lambda} e_{0,0} = 0$, but (2.19) degenerates to $0 = 0$, so it supplies no cancellation at the origin. Expanding Π_d therefore leaves only the origin:

$$\begin{aligned} \langle v, L_z H_{+,\lambda}\Pi_d w\rangle &= \langle v, L_z H_{+,\lambda} e_{0,0}\rangle\langle e_{0,0}, w\rangle, \\ \langle v, \Pi_d H_{+,\lambda} L_z w\rangle &= \langle v, e_{0,0}\rangle\langle L_z H_{+,\lambda} e_{0,0}, w\rangle. \end{aligned} \tag{2.21}$$

Subtracting the equations in (2.21) and substituting into (2.18) proves (2.17). □

3 Proof of Main Result

In Theorem 4.1, Lemma 2.4 is used to establish monotonicity of the maximum logarithmic height, defined below. We combine this with three external inputs to prove gap persistence up to criticality: one seed gap per label [22], the Thouless formula [38], and the Bourgain–Jitomirskaya formula for the logarithmic height function on the spectrum [11]. In what follows, assume $\lambda > 0$.

3.1 The logarithmic height

Definition 3.1 (Logarithmic potential)**.** For real E, define

$$\eta_\lambda(E) := \int \log|E - t|\,\mathrm{d}\nu_\lambda(t). \tag{3.1}$$

Theorem 3.2 (Thouless [38] and Bourgain–Jitomirskaya [11]; LE)**.** *The potential is finite and continuous in real energy, nonnegative, and satisfies*

$$\eta_\lambda(E) = \max\{0, \log\lambda\}, \qquad E \in \Sigma_\lambda. \tag{3.2}$$

Proof. The Thouless formula [38, 6], valid at every real energy by [13, Theorem 4.2], identifies $\eta_\lambda(E)$ with the one-step transfer matrix Lyapunov exponent of the associated almost Mathieu operator.

At each fixed real energy, the transfer matrices are uniformly bounded in phase and have determinant one. Their Lyapunov exponent, and hence $\eta_\lambda(E)$, is therefore finite and nonnegative.

Continuity in real energy follows from [11, Theorem 1]. Finally, [11, Corollary 2], with cosine amplitude 2λ in our notation, gives

$$\eta_\lambda(E) = \max\{0, \log\lambda\}, \qquad E \in \Sigma_\lambda. \qquad \square$$

Lemma 3.3 (Upper semicontinuity; L^*)**.** *The map $(\lambda, E) \mapsto \eta_\lambda(E)$ is upper semicontinuous on $(0,\infty)\times\mathbb{R}$.*

Proof. For $0 < \varepsilon \le 1$, set

$$F_\varepsilon(\lambda, E) := \tau_\alpha\big(\log(|E - H_\lambda| + \varepsilon I)\big) = \int_{\mathbb{R}} \log(|E-t| + \varepsilon)\, \mathrm{d}\nu_\lambda(t). \tag{3.3}$$

Norm continuity of H_λ and continuous functional calculus show that each F_ε is continuous in (λ, E). As ε decreases to zero, these integrals decrease, and

$$\lim_{\varepsilon\to 0^+} F_\varepsilon(\lambda, E) = \eta_\lambda(E). \tag{3.4}$$

Indeed, at each fixed (λ, E) the integrands decrease and are uniformly bounded above on the compact support of ν_λ, so monotone convergence applies after subtraction from a constant upper bound. Thus,

$$\eta_\lambda(E) = \inf_{0<\varepsilon\le 1} F_\varepsilon(\lambda, E), \tag{3.5}$$

showing upper semicontinuity. □

3.2 The maximum logarithmic height

Lemma 3.4 (Maximum height condition; L^*)**.** *Every bounded gap of label n has a unique maximizing energy $m_n(\lambda)$, characterized by*

$$\tau_\alpha(R_\lambda(m_n(\lambda))) = 0. \tag{3.6}$$

It depends locally real-analytically on λ, and $\eta_\lambda(m_n(\lambda)) > \max\{0, \log\lambda\}$.[4]

Proof. Differentiating (3.1) inside a gap gives

$$\partial_E \eta_\lambda(E) = \tau_\alpha(R_\lambda(E)) \quad \text{and} \quad \partial_E^2 \eta_\lambda(E) = -\tau_\alpha(R_\lambda(E)^2) = -\|R_\lambda(E) e_{0,0}\|^2 < 0. \tag{3.7}$$

The equal endpoint values from Theorem 3.2 and strict concavity give a unique interior maximum above that value. Joint resolvent analyticity and the nonzero second derivative give local real analyticity by the implicit-function theorem. □

Define the maximum height and its total derivative

$$\eta_n(\lambda) := \eta_\lambda(m_n(\lambda)), \qquad \eta_n'(\lambda) := \frac{\mathrm{d}}{\mathrm{d}\lambda}\big[\eta_\lambda(m_n(\lambda))\big]. \tag{3.8}$$

Lemma 3.5 (Maximum-height derivative; L^*)**.** *At every positive coupling where the bounded gap of label n is open, its maximum height satisfies*

$$\lambda\eta_n'(\lambda) = -\operatorname{Re}\langle R_\lambda(m_n(\lambda)) e_{0,0}, \lambda(\partial_\lambda H_{+,\lambda}) e_{0,0}\rangle. \tag{3.9}$$

[4]The characterization of maximum height by a vanishing averaged Green function was highlighted to D. S. B. by Qi Zhou in 2024. Riedel [33] considered an equivalent approach based on the logarithmic potential and its critical points.

Proof. Fix a gap energy E. The operator $(E-H_\lambda)^2$ remains uniformly positive for nearby couplings. Since $\eta_\lambda(E) = \frac{1}{2}\tau_\alpha\big(\log\big((E-H_\lambda)^2\big)\big)$, differentiating and using trace cyclicity yields

$$\partial_\lambda \eta_\lambda(E) = \frac{1}{2}\tau_\alpha\left(R_\lambda(E)^2\,\partial_\lambda[(E-H_\lambda)^2]\right) = -\tau_\alpha\left(R_\lambda(E)\partial_\lambda H_\lambda\right). \tag{3.10}$$

At $E = m_n(\lambda)$, $\partial_E\eta_\lambda(E) = 0$, so the chain rule gives

$$\eta_n'(\lambda) = \partial_\lambda\eta_\lambda(E) + m_n'(\lambda)\partial_E\eta_\lambda(E) = -\tau_\alpha(R_\lambda(E)\partial_\lambda H_\lambda). \tag{3.11}$$

Finally, the self-adjointness of $R_\lambda(E)$ and $(\partial_\lambda H_\lambda)e_{0,0} = (\partial_\lambda H_{+,\lambda})e_{0,0}$, gives

$$\lambda\eta_n'(\lambda) = -\operatorname{Re}\langle R_\lambda(E)e_{0,0}, \lambda(\partial_\lambda H_{+,\lambda})e_{0,0}\rangle, \tag{3.12}$$

as claimed. □

Theorem 3.6 (Monotonicity; L^*)**.** *Along every local branch of bounded gaps at positive coupling,*

$$0 \le \lambda\eta_n'(\lambda) \le 1. \tag{3.13}$$

Proof. This is Theorem 4.1. □

3.3 Continuation from a seed

A seed below the target coupling suffices.

Theorem 3.7 (Seed gaps [22, Théorème 2.2 and Proposition 2.5]; LE)**.** *For every irrational α and every $n \in \mathbb{Z}\setminus\{0\}$, the gap with IDS label $\{n\alpha\}$ is open at arbitrarily small positive couplings.*

Remark 3.8. Theorem 3.7 also follows from Liu–Yuan [29, Theorem 6.2], incorporating the result of Avila–Jitomirskaya [3, Theorem 8.2]: for each irrational α, there exists $\lambda_*(\alpha) > 0$ such that all admissible gaps are open whenever $0 < \lambda < \lambda_*(\alpha)$, yielding a seed-coupling threshold uniform in n.

Lemma 3.9 (Constancy of gap labels [8, Sec. 4.2, Theorem 16(iv)]; L^*)**.** *Along any continuous path of coupling–energy pairs (λ, E) with $E \notin \Sigma_\lambda$, the value $N_\lambda(E)$ is constant.*

Proof. Fix (λ_0, E_0) with $E_0 \notin \Sigma_{\lambda_0}$. Since, $\|H_\mu - H_\lambda\| \le 2|\mu-\lambda|$, a common Riesz contour Γ encloses precisely the spectrum below E for all nearby (λ, E). The Fermi projector is analytic in λ and independent of E within this neighborhood. Differentiating $\chi_{H_\lambda}(E)^2 = \chi_{H_\lambda}(E)$ gives $\chi_{H_\lambda}(E)\partial_\lambda\chi_{H_\lambda}(E)\chi_{H_\lambda}(E) = 0$. By trace cyclicity,

$$\partial_\lambda N_\lambda(E) = \tau_\alpha(\partial_\lambda\chi_{H_\lambda}(E)) = 2\tau_\alpha(\chi_{H_\lambda}(E)\partial_\lambda\chi_{H_\lambda}(E)) = 2\tau_\alpha(\chi_{H_\lambda}(E)\partial_\lambda\chi_{H_\lambda}(E)\chi_{H_\lambda}(E)) = 0. \tag{3.14}$$

Thus, the IDS is locally constant on the joint real resolvent set and along every continuous path. □

Theorem 3.10 (Persistence from seed; L^*)**.** *If a bounded labeled gap is open at $0 < \lambda_0 \le 1$, the same label is realized by a bounded open gap at every $\lambda \in [\lambda_0, 1]$.*

Proof. Let n be the label of the seed gap, and fix $\lambda_0 < \mu \le 1$. Consider all real resolvent pairs (λ, E) with $\lambda \in [\lambda_0, \mu]$, $\partial_E\eta_\lambda(E) = 0$, $N_\lambda(E) = \{n\alpha\}$, and $\eta_\lambda(E) \ge \eta_n(\lambda_0)$. The seed maximum makes this set nonempty. Since the label lies strictly between zero and one, E lies between spectral points, so $|E| \le \|H_\lambda\| \le 2 + 2\lambda \le 4$.

To prove closedness, let (λ_j, E_j) lie in this set and converge to (λ, E). Lemma 3.3 gives

$$\eta_\lambda(E) \geq \limsup_j \eta_{\lambda_j}(E_j) \geq \eta_n(\lambda_0) > 0. \tag{3.15}$$

As $\lambda \leq 1$, Theorem 3.2 places E off the spectrum. Resolvent continuity and Lemma 3.9 give $\partial_E \eta_\lambda(E) = 0$ and $N_\lambda(E) = \{n\alpha\}$, proving closedness.

It follows that the set is compact, and thus its coupling coordinate has a maximum. If that maximum were less than μ, the local branch from Lemma 3.4, label constancy, and Theorem 3.6 would give a point farther to the right in the same set, a contradiction. Thus, the set contains a pair at coupling μ. □

Completion of the proof of Theorem 1.5. For $0 < \lambda \leq 1$, choose a seed $0 < \lambda_0 < \lambda$ for each $n \neq 0$ via Theorem 3.7; Theorem 3.10 then gives the labeled gap at λ.

The conjugations (1.6) and (1.7) are unitary and fix the origin, so they preserve spectra and the origin spectral measure. Thus, for $\lambda > 0$,

$$\Sigma_\lambda = \lambda \Sigma_{1/\lambda}, \qquad N_\lambda(E) = N_{1/\lambda}(E/\lambda), \tag{3.16}$$

and for $\lambda < 0$, $\Sigma_\lambda = \Sigma_{|\lambda|}$ and $N_\lambda = N_{|\lambda|}$. For $\lambda > 1$, scale the gap endpoints at $1/\lambda$ by λ; for $\lambda < 0$, take those at $|\lambda|$. In both cases, the label is unchanged and endpoints belong to the spectrum. □

4 Proof of Monotonicity

The sector parity L_z falls out of the height function derivative by restoring the horizontal hopping phase k_x in the AMO fiber,

$$(h_\lambda(k_x, k_y)\psi)_j = e^{\mathrm{i}k_x}\psi_{j+1} + e^{-\mathrm{i}k_x}\psi_{j-1} + 2\lambda\cos(\Theta j + k_y)\psi_j, \tag{4.1}$$

and differentiating the two momentum variables:

$$2\lambda\partial_\lambda h_\lambda(k_x, k_y) = (\partial_{k_x}^2 - \partial_{k_y}^2 + 1)h_\lambda(k_x, k_y). \tag{4.2}$$

In this sense, we are treating λ as originating from an analytic continuation of the momentum balanced between k_x and k_y. We recover the sector parity L_z in (2.16) by Fourier transforming and noting that h_λ only has nearest-neighbor hopping. Then, (4.2) gives, by stationarity,

$$\lambda(\partial_\lambda H_{+,\lambda})e_{0,0} = \frac{1}{2}(I - L_z)H_{+,\lambda}e_{0,0}. \tag{4.3}$$

Theorem 4.1 (Monotonicity; L^*)**.** *Along every local branch of bounded gaps at positive coupling,*

$$\lambda\eta_n'(\lambda) = \frac{\langle R_\lambda(m_n(\lambda))e_{0,0}, (I - L_z)R_\lambda(m_n(\lambda))e_{0,0}\rangle}{2\|R_\lambda(m_n(\lambda))e_{0,0}\|^2} \in [0, 1]. \tag{4.4}$$

Proof. Set $E = m_n(\lambda)$. Substitute the sector parity for the coupling derivative (4.3) into the maximum logarithmic height derivative (3.9) and simplify using $\tau_\alpha(R_\lambda(E)) = 0$ and the resolvent equation $\langle R_\lambda(E)e_{0,0}, H_{+,\lambda}e_{0,0}\rangle = -1$:

$$\lambda\eta_n'(\lambda) = \frac{1}{2} + \frac{1}{2}\,\mathrm{Re}\langle e_{0,0}, H_{+,\lambda}L_z R_\lambda(E)e_{0,0}\rangle. \tag{4.5}$$

Now, for a test vector $v \in \mathcal{V}_\lambda$, apply Lemma 2.4 with $w = R_\lambda(E)e_{0,0} \in \mathcal{V}_\lambda$. Since $\langle e_{0,0}, R_\lambda(E)e_{0,0}\rangle = 0$, this yields

$$\langle v, [L_z, H_{+,\lambda}]R_\lambda(E)e_{0,0}\rangle = -\langle v, e_{0,0}\rangle\langle e_{0,0}, H_{+,\lambda}L_zR_\lambda(E)e_{0,0}\rangle. \tag{4.6}$$

Observing that the right-hand-side of (4.6) is the term in (4.5), we expand the left-hand-side by applying the resolvent equation and $L_ze_{0,0} = 0$ to get

$$\begin{aligned}\langle v, [L_z, H_{+,\lambda}]R_\lambda(E)e_{0,0}\rangle &= \langle v, L_zH_{+,\lambda}R_\lambda(E)e_{0,0} - H_{+,\lambda}L_zR_\lambda(E)e_{0,0}\rangle &(4.7)\\ &= \langle v, L_z(ER_\lambda(E)e_{0,0} - e_{0,0}) - H_{+,\lambda}L_zR_\lambda(E)e_{0,0}\rangle &(4.8)\\ &= \langle (E - H_{+,\lambda})v, L_zR_\lambda(E)e_{0,0}\rangle. &(4.9)\end{aligned}$$

The last step uses $L_ze_{0,0} = 0$ and self-adjointness of $E - H_{+,\lambda}$.

This motivates the choice $v = R_\lambda(E)^2e_{0,0} \in \mathcal{V}_\lambda$ such that $(E - H_{+,\lambda})v = R_\lambda(E)e_{0,0}$. Since, $\langle v, e_{0,0}\rangle = \|R_\lambda(E)e_{0,0}\|^2 > 0$, we can rearrange (4.6) and substitute:

$$\langle e_{0,0}, H_{+,\lambda}L_zR_\lambda(E)e_{0,0}\rangle = -\frac{\langle v, [L_z, H_{+,\lambda}]R_\lambda(E)e_{0,0}\rangle}{\langle v, e_{0,0}\rangle} = -\frac{\langle R_\lambda(E)e_{0,0}, L_zR_\lambda(E)e_{0,0}\rangle}{\|R_\lambda(E)e_{0,0}\|^2}. \tag{4.10}$$

Finally, substituting back into (4.5) yields

$$\lambda\eta_n'(\lambda) = \frac{1}{2}\left(1 - \frac{\langle R_\lambda(E)e_{0,0}, L_zR_\lambda(E)e_{0,0}\rangle}{\|R_\lambda(E)e_{0,0}\|^2}\right). \tag{4.11}$$

Since $0 \le \|L_z\| \le 1$, this proves (4.4) and Theorem 3.6. □

Corollary 4.2 (Critical height; L^*)**.** *For every open bounded gap at critical coupling, $\eta_n'(1) = 1/2$.*

Proof. Origin-preserving duality and uniqueness of the gap maximum give

$$m_n(\lambda) = \lambda m_n(1/\lambda), \qquad \eta_n(\lambda) = \log\lambda + \eta_n(1/\lambda). \tag{4.12}$$

Differentiating at $\lambda = 1$ gives $\eta_n'(1) = 1 - \eta_n'(1)$. This uses the local branch of an open gap; it supplies no seed at criticality. □

4.1 Conserved SU(2) subspace

We now express duality as a spin symmetry on a two-dimensional space associated with each gap maximum. Fix $\lambda > 0$, and let P_λ be the orthogonal projection onto $\mathcal{V}_\lambda \cap e_{0,0}^\perp$. On $\operatorname{ran} P_\lambda$, define the centered Hamiltonian and angular observable by

$$H_{+,\lambda}^0 = P_\lambda H_{+,\lambda}P_\lambda, \qquad L_{z,\lambda}^0 = P_\lambda L_zP_\lambda. \tag{4.13}$$

Pair this space with its dual and, on $\operatorname{ran} P_\lambda \oplus \operatorname{ran} P_{1/\lambda}$, set

$$\mathbb{H}_\lambda^0 = H_{+,\lambda}^0 \oplus \lambda H_{+,1/\lambda}^0, \qquad \mathbb{L}_z^0 = L_{z,\lambda}^0 \oplus L_{z,1/\lambda}^0. \tag{4.14}$$

Then, for a bounded gap of label n, normalize its resolvent source:

$$u_n = R_\lambda(m_n(\lambda))e_{0,0}\|R_\lambda(m_n(\lambda))e_{0,0}\|^{-1}. \tag{4.15}$$

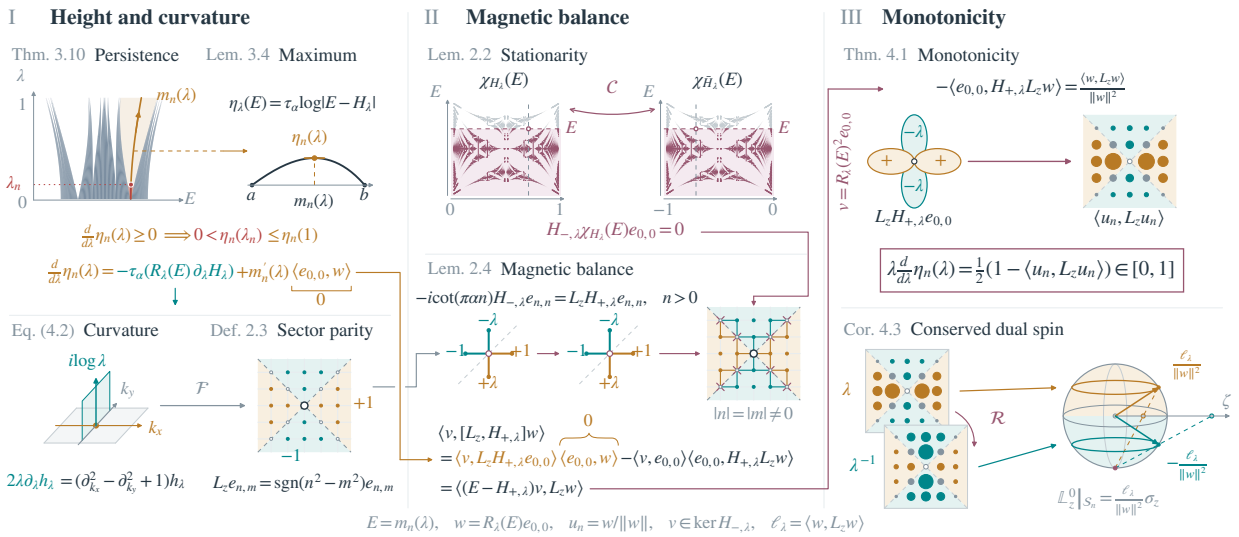


Figure 1: **I.** Selecting the gap maximum removes the energy term in the total derivative (gold). The remaining term (teal) is expressed via the sector parity L_z. **II.** Magnetic inversion pairs the Fermi-projector sources at the origin, giving stationarity. This cancels nonzero diagonal pairings, and magnetic balance leaves the origin terms. **III.** Choosing $v = R_\lambda(E)^2 e_{0,0}$ identifies the surviving origin response with the normalized L_z, yielding the required $0 \le \lambda\eta_n'(\lambda) \le 1$ to keep the (red) seed gap open up to $\lambda = 1$. Below, quarter-turn duality gives the conserved spin doublet. References identify the proof steps, arrows indicate dependencies, and the coupling spectrum uses $\alpha \simeq \sqrt{2}/2$.

Lemma 3.4 gives $\langle e_{0,0}, u_n\rangle = 0$, so $u_n \in \operatorname{ran} P_\lambda$. The quarter-turn supplies the dual state $\mathcal{R}u_n \in \operatorname{ran} P_{1/\lambda}$. These states define the dual doublet

$$\mathcal{S}_n = \operatorname{span}\{u_n \oplus 0,\ 0 \oplus \mathcal{R}u_n\}. \tag{4.16}$$

In this ordered orthonormal basis, use the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.17}$$

Corollary 4.3 (Conserved dual spin; L^*)**.** *The doublet $\mathcal{S}_n$ reduces both paired operators, with*

$$\mathbb{H}^0_\lambda|_{\mathcal{S}_n} = m_n(\lambda) I_2, \qquad \mathbb{L}^0_z|_{\mathcal{S}_n} = (1 - 2\lambda\eta_n'(\lambda))\sigma_z. \tag{4.18}$$

Consequently, the generators $\sigma_j/2$ define a conserved spin-$\frac{1}{2}$ SU(2) action on $\mathcal{S}_n$. Its compressed, time-reversal-even angular polarization selects the z-axis when nonzero.

Proof. Since $\mathcal{V}_\lambda$ reduces $H_{+,\lambda}$ and $L_z e_{0,0} = 0$, magnetic balance gives

$$[L^0_{z,\lambda}, H^0_{+,\lambda}] = P_\lambda [L_z, H_{+,\lambda}] P_\lambda = 0. \tag{4.19}$$

The resolvent equation and Lemma 3.4 give $\ker(m_n(\lambda) - H^0_{+,\lambda}) = \mathbb{C}u_n$: a vector v in this kernel satisfies $(m_n(\lambda) - H_{+,\lambda})v \in \mathbb{C}e_{0,0}$, and (2.12) makes this operator invertible on $\mathcal{V}_\lambda$. Commutation and (4.11) imply $L^0_{z,\lambda} u_n = (1 - 2\lambda\eta_n'(\lambda))u_n$. The quarter-turn maps P_λ to $P_{1/\lambda}$ and gives

$$\mathcal{R} H^0_{+,\lambda} \mathcal{R}^* = \lambda H^0_{+,1/\lambda}, \qquad \mathcal{R} L^0_{z,\lambda} \mathcal{R}^* = -L^0_{z,1/\lambda}. \tag{4.20}$$

Hence both restrictions hold, and the Pauli generators commute with the restricted Hamiltonian. □

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Authors' Note This manuscript revisits the approach in arXiv: 2112.06869v1, whose original argument did not establish the claimed theorem. Further development of the projected Green's function and transfer matrix formalism motivated the magnetic conjugation and diagonal boundary balance used here.

After publishing our work, we were made aware by Qi Zhou that Lemma 4.3 in arXiv:2112.06869v2 motivates an orbit multiplicity argument in the dynamical systems language and extends the formalism introduced in [5], further highlighting the scope of the symmetry discussed in Section 2.

A Proof correspondence and verification

The Lean formalization for this project can be found on M. F.'s github:
https://mattfaust.github.io/Dry10/dtm_LEAN.zip.

A.1 The complete statement, in parts

Every theorem-specific definition is written below, in the order in which it is used. Parts 1–5 form `DTMAudit/Submission/Definitions.lean`, which imports only Mathlib. Part 6 is `DTMAudit/Submission/Theorem.lean`; it imports those definitions and the current compiled proof. Thus the final theorem is short, but none of its mathematical definitions is left implicit.

`lam` denotes λ, and the phase is measured in radians. `SequenceSpace` is $\ell^2(\mathbb{Z};\mathbb{C})$, and `→L[ℂ]` denotes bounded complex-linear maps. The family `h` consists of the fiber AMO operators; the manuscript's uppercase H_λ remains the magnetic lattice operator. In operator products, `1` is the identity and • is scalar multiplication.

1. Frequency and sequence space

```
import Mathlib.Analysis.InnerProductSpace.l2Space
import Mathlib.Algebra.Algebra.Spectrum.Basic
import Mathlib.LinearAlgebra.Matrix.Charpoly.Basic
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic
import Mathlib.Data.Real.Irrational

/- Every theorem-specific definition is printed here, in dependency order.
   Only Mathlib is imported. The phase is measured in radians.
   ‘lam‘ denotes the coupling lambda; lambda itself is Lean syntax. -/

noncomputable section
namespace DryTenMartini

-- 1. Frequency and the Hilbert space of square-summable sequences.

def IrrationalFrequency (α : ℝ) : Prop :=
  ∀ p : ℤ, ∀ q : ℕ, q ≠ 0 → (p : ℝ) / (q : ℝ) ≠ α

abbrev SequenceSpace := lp (fun _ : ℤ => ℂ) 2
```

2. The operator and its inverse

```
-- 2. The almost Mathieu operator and its bounded inverse.
-- ‘→L[ℂ]‘ means a bounded complex-linear operator.

def AMOFamily (α : ℝ)
    (h : ℝ → ℝ → (SequenceSpace →L[ℂ] SequenceSpace)) : Prop :=
  ∀ (lam φ : ℝ) (ψ : SequenceSpace) (j : ℤ),
    h lam φ ψ j =
      ψ (j + 1) + ψ (j - 1) +
      ((2 * lam * Real.cos
        (2 * Real.pi * α * (j : ℝ) + φ) : ℝ) : ℂ) * ψ j

-- Multiplication of operators is composition; ‘1‘ is the identity.
-- Its negation means that E belongs to the operator spectrum.
def InvertibleAt (h : SequenceSpace →L[ℂ] SequenceSpace)
```

```
    (E : ℝ) : Prop :=
  ∃ inverse : SequenceSpace →L[ℂ] SequenceSpace,
    ((E : ℂ) • 1 - h) * inverse = 1 ∧
    inverse * ((E : ℂ) • 1 - h) = 1
```

3. Finite Dirichlet eigenvalue counts

```
-- 3. Dirichlet eigenvalue counts on the sites 0,...,L.

def DirichletMatrix (α lam φ : ℝ) (L : ℕ) :
    Matrix (Fin (L + 1)) (Fin (L + 1)) ℂ :=
  fun i j =>
    (if (i.val : ℤ) + 1 = (j.val : ℤ) then (1 : ℂ) else 0) +
    (if (i.val : ℤ) - 1 = (j.val : ℤ) then (1 : ℂ) else 0) +
    (if i = j then
      ((2 * lam * Real.cos
        (2 * Real.pi * α * (i.val : ℝ) + φ) : ℝ) : ℂ)
     else 0)

-- The matrix is Hermitian, so its eigenvalues are real.
-- The roots form a multiset: countP includes algebraic multiplicities.
def NormalizedCount (α lam φ E : ℝ) (L : ℕ) : ℝ :=
  ((((Matrix.charpoly (DirichletMatrix α lam φ L)).roots.countP
      (fun z : ℂ => z.re < E) : ℕ) : ℝ) / ((L + 1 : ℕ) : ℝ))
```

4. The limiting gap label

```
-- 4. The gap label, written as an epsilon-cutoff limit.
-- The cutoff can depend on the phase and energy: no uniformity is assumed.

def HasGapLabel (α lam φ : ℝ) (n : ℤ) (E : ℝ) : Prop :=
  ∀ ε : ℝ, 0 < ε →
    ∃ L0 : ℕ, ∀ L : ℕ, L0 ≤ L →
      |NormalizedCount α lam φ E L -
        ((n : ℝ) * α - (Int.floor ((n : ℝ) * α) : ℝ))| < ε
```

5. The common labelled gap

```
-- 5. A common open spectral gap with this label at every phase.
-- Its endpoints are in the spectrum, and every interior energy is out.

def CommonLabelledGap (α : ℝ)
    (h : ℝ → ℝ → (SequenceSpace →L[ℂ] SequenceSpace))
    (lam : ℝ) (n : ℤ) (a b : ℝ) : Prop :=
  a < b ∧ ∀ φ : ℝ,
    ¬ InvertibleAt (h lam φ) a ∧
    ¬ InvertibleAt (h lam φ) b ∧
    ∀ E : ℝ, a < E → E < b →
      InvertibleAt (h lam φ) E ∧ HasGapLabel α lam φ n E

end DryTenMartini
```

6. The final theorem

```
import DTMAudit.Submission.Definitions
import DTMAudit.CurrentDraft

namespace DryTenMartini

-- 6. Every allowed nonzero label is an open gap, at every nonzero coupling.
-- The same endpoints work for all phases: they are chosen in the definition
-- before the universal phase quantifier.

theorem dry_ten_martini :
    ∀ α : ℝ, IrrationalFrequency α →
      ∃ h : ℝ → ℝ → (SequenceSpace →L[ℂ] SequenceSpace),
        AMOFamily α h ∧
        ∀ lam : ℝ, lam ≠ 0 → ∀ n : ℤ, n ≠ 0 →
          ∃ a b : ℝ, CommonLabelledGap α h lam n a b := by
  have hirr (α : ℝ) : Irrational α ↔
      ∀ p : ℤ, ∀ q : ℕ, q ≠ 0 → (p : ℝ) / (q : ℝ) ≠ α := by
    constructor
    · intro h p q _ heq
      apply ((irrational_iff_ne_rational α).mp h p (q : ℤ))
      simpa only [Int.cast_natCast] using heq.symm
    · intro h
      rw [Irrational, Set.mem_range]
      rintro ⟨q, hq⟩
      exact h q.num q.den q.den_nz
        (by simpa only [Rat.cast_def] using hq)
  simpa only [IrrationalFrequency, AMOFamily, CommonLabelledGap,
    InvertibleAt, HasGapLabel, NormalizedCount, DirichletMatrix,
    hirr, not_exists, spectrum.not_mem_iff, spectrum.mem_iff,
    isUnit_iff_exists, Algebra.algebraMap_eq_smul_one,
    Metric.tendsto_atTop, Real.dist_eq, Int.fract,
    not_forall, not_not] using CurrentDraft.dry_ten_martini

end DryTenMartini
```

The finite matrix is the Dirichlet restriction to $0, \dots, L$; its roots are counted with algebraic multiplicity and divided by $L + 1$. The matrix is Hermitian, so its eigenvalues are real. At a gap energy, the strict count agrees with the closed Fermi-projector convention in (1.10). The same $a < b$ work for every phase, because the phase quantifier occurs inside `CommonLabelledGap`. The convergence cutoff may depend on phase and energy.

Unfolding all eight definitions gives the proposition in `DTMAudit/Submission/ExpandedStatement.lean` definitionally. The theorem `DryTenMartini.exact_unfolding` in `DTMAudit/Submission/Unfolding.lean` proves this with `Iff.rfl`. The check in `audit/SubmissionSignatureAudit.lean` also verifies that `DryTenMartini.dry_ten_martini` has exactly this expanded type and that its mathematical constants come only from Lean's standard library and Mathlib. The six printed code blocks are generated from their compiled sources; joining Parts 1–5 recovers the definitions file exactly after reversing the typesetting substitutions.

Remark A.1. No operator-model, seed, or monotonicity hypothesis is added. A formalization of existing literature for seed gaps was constructed and used instead.

A.2 Correspondence with the written proof

The checked proofs follow the arguments associated with the manuscript's L^* labels. Lemma 2.2 uses the commuting magnetic actions and their common action at the origin to obtain stationarity on $\mathcal{V}_\lambda$. Lemma 2.4 reduces the diagonal boundary pairings to the origin contribution. Spectral calculus places $R_\lambda(E)e_{0,0}$ and $R_\lambda(E)^2e_{0,0}$ in $\mathcal{V}_\lambda$; invariance of this space under L_z is not required.

For Lemma 3.3, decreasing continuous logarithmic regularizations give upper semicontinuity. Strict concavity and the implicit-function theorem give $m_n(\lambda)$ in Lemma 3.4. The proof of Lemma 3.5 differentiates $\frac{1}{2}\tau_\alpha(\log((E-H_\lambda)^2))$ where the positive operator is locally uniformly bounded below, combines the product-rule terms by trace cyclicity, and applies the chain rule at the maximum. The source vector $e_{0,0}$ remains fixed.

Theorem 4.1 combines this derivative with magnetic balance and the origin calibration of L_z to obtain (4.4), and hence Theorem 3.6. In Lemma 3.9, norm differentiation of the Riesz projector gives $\chi_{H_\lambda}(E)\chi'_{H_\lambda}(E)\chi_{H_\lambda}(E) = 0$ and $\tau_\alpha(\chi'_{H_\lambda}(E)) = 0$. These results enter the compact stationary superlevel argument of Theorem 3.10. Theorems 3.2 and 3.7, followed by Aubry scaling and the sign involution, complete Theorem 1.5.

Corollary 4.2 follows by differentiating height duality at an open critical gap. The checked argument for Corollary 4.3 uses the centered compression, the resolvent-source eigenspace, and the quarter-turn to construct the conserved dual spin. These corollaries are separate consequences of the preceding identities.

A.3 External inputs

The *LE* labels on Theorems 3.2 and 3.7 indicate that the formalization proves the required conclusions for the actual almost Mathieu operator and its integrated density of states, but does not reproduce the cited historical proofs in full. The checked normalization includes logarithmic integrability; the checked seed result supplies arbitrarily small positive couplings for each nonzero label. These inputs are discharged in the proof of Theorem 1.5. Dependencies retain their own verification labels.

A.4 Verification Scope

The verification records are `audit/latest-source-build.json` and `audit/latest-verification.json`. A fresh source build and kernel replay passed for all 223 project modules and 3,612 safe logical declarations, with none skipped and no custom project axioms. The only logical axioms used were `propext`, `Classical.choice`, and `Quot.sound`. The modular theorem, expanded statement, and definitional unfolding are included in this verification. Exact manuscript-to-source correspondence is recorded in `CORRESPONDENCE.json`.

B Alternative Monotonicity Proof

In this section, we give an alternative proof of height monotonicity, using the height-derivative formula from Section 3. This proof uses the proof strategies of the previous version of our manuscript, arXiv:2112.06869v2. We repeat several definitions for the sake of self-containment. The following table relates the results of this appendix to those in Sections 2 and 4.

Appendix result	**Main-text counterpart**	**Relationship**
Lemma B.1	Equation (2.9) in Lemma 2.2	The same commutation and common-origin identities, computed explicitly in the Landau basis.
Lemma B.2	Equation (2.19) in Lemma 2.4	The same balance at nonzero diagonal sites.
Lemma B.3	The boundary cancellation in Lemma 2.4	Packages this cancellation into the operator identity $QHu = Ku$ for stationary vectors when $u_{0,0} = 0$.
Lemma B.4	Equations (4.6) and (4.7) in Theorem 4.1	Provides an alternative resolvent identity using cone compression.
Theorem B.5	Theorem 4.1	Obtains the same normalized angular response and the bounds $0 \le \lambda\eta_n'(\lambda) \le 1$.

Table 1: Correspondence between the cone-compression proof and Sections 2 and 4. The intermediate results organize the same diagonal balance in different ways.

B.1 Magnetic conjugation and setup

Fix an irrational α and a coupling $\lambda > 0$, and put $\Theta = 2\pi\alpha$. We work on $\ell^2(\mathbb{Z}^2)$ in the Landau basis $\tilde{e}_{n,m}$. Write $\Omega = \tilde{e}_{0,0} = e_{0,0}$ and $u_{n,m} = \langle \tilde{e}_{n,m}, u\rangle$. Recall the magnetic translations and Hamiltonian:

$$U\tilde{e}_{n,m} = \tilde{e}_{n-1,m}, \qquad V\tilde{e}_{n,m} = e^{\mathrm{i}\Theta n}\tilde{e}_{n,m-1}, \qquad H = H_\lambda = U + U^* + \lambda(V + V^*). \tag{B.1}$$

The spectrum is Σ_λ, and the resolvent is $R_\lambda(E) = (E - H_\lambda)^{-1}$ for $E \notin \Sigma_\lambda$.

Magnetic conjugation is given by

$$(\mathcal{C}u)_{n,m} = e^{-\mathrm{i}\Theta nm}\overline{u_{n,m}}. \tag{B.2}$$

Set

$$\bar{H} = \bar{H}_\lambda = \mathcal{C}H_\lambda\mathcal{C}^*, \qquad H_+ = H_{+,\lambda} = \tfrac{1}{2}(H_\lambda + \bar{H}_\lambda). \tag{B.3}$$

Lemma B.1. *The operators H and $\bar{H}$ commute and agree on the origin vector:*

$$[H, \bar{H}] = 0, \qquad H\Omega = \bar{H}\Omega = \tilde{e}_{-1,0} + \tilde{e}_{1,0} + \lambda(\tilde{e}_{0,-1} + \tilde{e}_{0,1}). \tag{B.4}$$

Proof. Applying (B.2) on both sides of each translation gives

$$\mathcal{C}U\mathcal{C}^*\,\tilde{e}_{n,m} = e^{\mathrm{i}\Theta m}\tilde{e}_{n-1,m}, \qquad \mathcal{C}V\mathcal{C}^*\,\tilde{e}_{n,m} = \tilde{e}_{n,m-1}. \tag{B.5}$$

The shift U commutes with both conjugated shifts, and V commutes with $\mathcal{C}V\mathcal{C}^*$, since the corresponding phases are unchanged by these shifts. For the remaining pair, the two orders give

$$V(\mathcal{C}U\mathcal{C}^*)\tilde{e}_{n,m} = (\mathcal{C}U\mathcal{C}^*)V\tilde{e}_{n,m} = e^{\mathrm{i}\Theta(n+m-1)}\tilde{e}_{n-1,m-1}. \tag{B.6}$$

All four shifts are unitary, so these commutations also hold with their adjoints. Expanding H and $\bar{H}$ now gives $[H, \bar{H}] = 0$. At the origin, all phase factors in the shift formulas are one, giving the displayed common action. □

For real $E \notin \Sigma_\lambda$, set

$$g(E) = \langle \Omega, R_\lambda(E)\Omega \rangle, \qquad q(E) = R_\lambda(E)\Omega - g(E)\Omega. \tag{B.7}$$

Thus $q(E)$ is the resolvent column with its origin coefficient removed. By Lemma B.1, $H - \bar{H}$ commutes with $E - H$ and hence with its inverse. Therefore

$$(H - \bar{H})R_\lambda(E)\Omega = R_\lambda(E)(H - \bar{H})\Omega = 0. \tag{B.8}$$

The same difference annihilates $g(E)\Omega$, so

$$Hq(E) = \bar{H}q(E), \qquad q(E)_{0,0} = 0. \tag{B.9}$$

B.2 Cone compression

Let

$$\Gamma = \{(n,m) : |m| = |n|\} \setminus \{(0,0)\}, \qquad S = \{(n,m) : |m| \geq |n|\} \setminus \{(0,0)\}, \tag{B.10}$$

where all coordinates are integers. Let P_S be the orthogonal projection that keeps the coefficients in S and sets all other coefficients to zero. Define the weight Q and the compressed operator K by

$$(Qu)_{n,m} = \begin{cases} u_{n,m}, & |m| > |n|, \\ \frac{1}{2}u_{n,m}, & (n,m) \in \Gamma, \\ 0, & \text{otherwise}, \end{cases} \qquad K = P_S H_+ P_S. \tag{B.11}$$

Note that K keeps precisely those hopping terms whose two endpoints lie in S. The half-weight in Q will account for the two hopping terms removed at each nonzero diagonal site.

For clarity, we record the Landau-coordinate formulas for the two actions.[5]

$$(Hu)_{n,m} = u_{n+1,m} + u_{n-1,m} + \lambda e^{\mathrm{i}\Theta n}u_{n,m+1} + \lambda e^{-\mathrm{i}\Theta n}u_{n,m-1}, \tag{B.12}$$

$$(\bar{H}u)_{n,m} = e^{\mathrm{i}\Theta m}u_{n+1,m} + e^{-\mathrm{i}\Theta m}u_{n-1,m} + \lambda u_{n,m+1} + \lambda u_{n,m-1}. \tag{B.13}$$

With $t_r = (1 + e^{\mathrm{i}\Theta r})/2 = e^{\mathrm{i}\pi\alpha r}\cos(\pi\alpha r)$, their average is

$$(H_+u)_{n,m} = t_m u_{n+1,m} + \overline{t_m}u_{n-1,m} + \lambda t_n u_{n,m+1} + \lambda\overline{t_n}u_{n,m-1}. \tag{B.14}$$

Lemma B.2. *Suppose $Hu = \bar{H}u$. At every site $(n,m) \in \Gamma$, the sum in* (B.14) *coming from the strict vertical cones equals the sum coming from the strict horizontal cones. That is, $(Ku)_{n,m} = \frac{1}{2}(Hu)_{n,m}$.*

Proof. Subtracting (B.13) from (B.12) gives

$$\sin(\pi\alpha m)\big(e^{-\mathrm{i}\pi\alpha m}u_{n-1,m} - e^{\mathrm{i}\pi\alpha m}u_{n+1,m}\big) = \lambda\sin(\pi\alpha n)\big(e^{-\mathrm{i}\pi\alpha n}u_{n,m-1} - e^{\mathrm{i}\pi\alpha n}u_{n,m+1}\big). \tag{B.15}$$

Fix an integer $r \neq 0$. Irrationality of α gives $\sin(\pi\alpha r) \neq 0$. At $(n,m) = (r,r)$, divide by this sine, rearrange, and multiply by $\cos(\pi\alpha r)$ to obtain

$$t_r u_{r+1,r} + \lambda\overline{t_r}u_{r,r-1} = \overline{t_r}u_{r-1,r} + \lambda t_r u_{r,r+1}. \tag{B.16}$$

At $(n,m) = (r,-r)$, the same calculation gives

$$\overline{t_r}u_{r+1,-r} + \lambda t_r u_{r,-r+1} = t_r u_{r-1,-r} + \lambda\overline{t_r}u_{r,-r-1}. \tag{B.17}$$

For $r > 0$, the left side of each identity comes from the horizontal cones and the right side from the vertical cones. For $r < 0$, these roles reverse. In particular, $(H_+u)_{r,\pm r} = 2(Ku)_{r,\pm r}$. As $Hu = \bar{H}u$ we have that $H_+u = Hu$ and so $(Ku)_{r,\pm r} = \frac{1}{2}(Hu)_{r,\pm r}$. □

[5]The corresponding stationary lattice equations, together with their degeneracy along the diagonals $n = \pm m$, also appear in Riedel [33, Section 2, equation (2.1)].

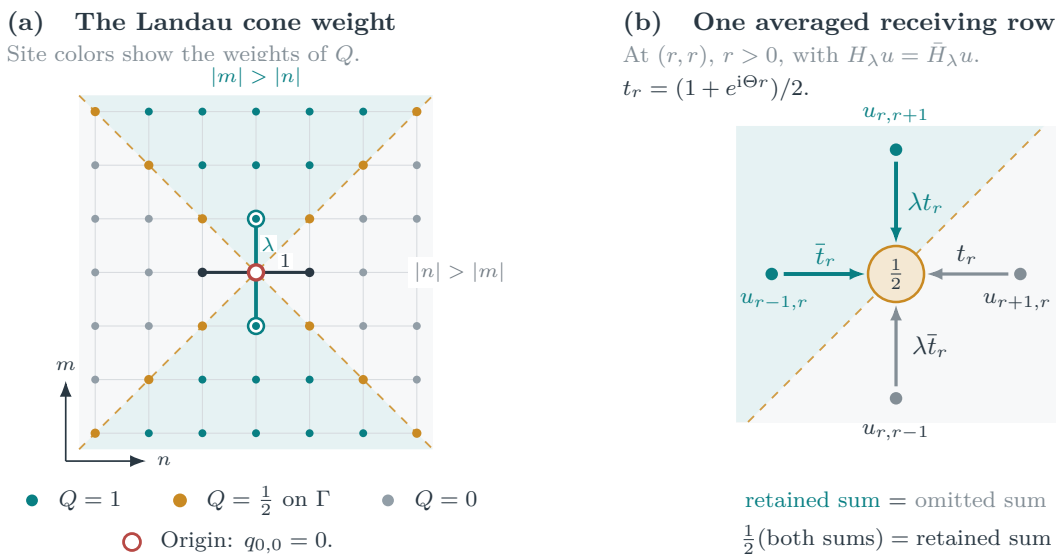


Figure 2: Left: Q has weight one in the strict vertical cones, one-half on Γ, and zero elsewhere. The hollow origin records $q(E)_{0,0} = 0$; the vertical neighbors support $QH\Omega$. Right: at (r, r) with $r > 0$, compressing H_+ retains the two contributions from the vertical cone. Lemma B.2 shows that their sum equals the sum of the two omitted contributions, explaining the half-weight on Γ.

Lemma B.3. *If $u_{0,0} = 0$ and $Hu = \bar{H}u$, then*

$$QHu = Ku. \tag{B.18}$$

Proof. The hypothesis gives $H_+u = Hu$. At a strict vertical site, no nearest neighbor is in a strict horizontal cone. Restricting the input to S therefore removes at most the origin term, which is zero. Thus $(Ku)_{n,m} = (Hu)_{n,m}$, matching the weight one in Q.

At a nonzero diagonal site, Lemma B.2 gives $(Ku)_{n,m} = \frac{1}{2}(Hu)_{n,m}$, matching the weight $\frac{1}{2}$ in Q.

Outside S, including the origin (by hypothesis $u_{0,0} = 0$), QHu vanishes by its weight and Ku vanishes by the output projection P_S. This proves the equality at every site. □

B.3 The resolvent identity and the height derivative

Define the source vector

$$f_\lambda = QH\Omega = \lambda(\tilde{e}_{0,1} + \tilde{e}_{0,-1}) = \lambda(\partial_\lambda H_{+,\lambda})\Omega. \tag{B.19}$$

The last equality follows by differentiating the common origin action (B.4) in λ. Since $\Omega = e_{0,0}$, this is exactly the source in the height-derivative formula (3.9).

Lemma B.4. *For every real $E \notin \Sigma_\lambda$,*

$$(EQ - K)q(E) = g(E)f_\lambda. \tag{B.20}$$

Proof. The resolvent equation and (B.7) give

$$(E - H)q(E) = (1 - Eg(E))\Omega + g(E)H\Omega. \tag{B.21}$$

Apply Q. The first term on the right vanishes because $Q\Omega = 0$, and the second becomes $g(E)f_\lambda$. By (B.9), Lemma B.3 applies to $q(E)$ and replaces $QHq(E)$ by $Kq(E)$. □

Theorem B.5. *Along every local branch of bounded gaps at positive coupling, the maximum height satisfies*

$$0 \le \lambda\eta_n'(\lambda) \le 1. \tag{B.22}$$

In particular, the maximum height is nondecreasing in λ.

Proof. Keep λ fixed and first differentiate (B.20) with respect to E:

$$Qq(E) + (EQ - K)\partial_E q(E) = (\partial_E g(E))f_\lambda. \tag{B.23}$$

The differentiation is valid in norm because the resolvent is analytic off the spectrum. Now set $m = m_n(\lambda)$ and $q = q(m)$. Lemma 3.4 gives $g(m) = 0$, so

$$q = R_\lambda(m)\Omega \neq 0, \qquad (mQ - K)q = 0. \tag{B.24}$$

with the second equality by Lemma B.4. Take the inner product of (B.23) at $E = m$ with q:

$$\langle q, Qq\rangle + \langle q, (mQ - K)\partial_E q(m)\rangle = (\partial_E g(m))\langle q, f_\lambda\rangle. \tag{B.25}$$

As $mQ - K$ is self-adjoint, by (B.24),

$$\langle q, (mQ - K)\partial_E q(m)\rangle = \langle (mQ - K)q, \partial_E q(m)\rangle = 0. \tag{B.26}$$

Thus, by (B.25) and (B.26),

$$\langle q, Qq\rangle = (\partial_E g(m))\langle q, f_\lambda\rangle. \tag{B.27}$$

Equations (3.7) and (3.9) from Section 3 give

$$\partial_E g(m) = -\|q\|^2, \qquad \lambda\eta_n'(\lambda) = -\operatorname{Re}\langle q, f_\lambda\rangle. \tag{B.28}$$

Taking real parts of the paired equation and substituting these identities yields

$$\langle q, Qq\rangle = -\|q\|^2 \operatorname{Re}\langle q, f_\lambda\rangle = \|q\|^2\lambda\eta_n'(\lambda). \tag{B.29}$$

Dividing by $\|q\|^2 > 0$, we obtain

$$\lambda\eta_n'(\lambda) = \frac{\langle q, Qq\rangle}{\|q\|^2} \in [0, 1], \tag{B.30}$$

where the last inclusion follows because the diagonal weights of Q belong to $\{0, \frac{1}{2}, 1\}$. □

Remark B.6. In the convention of the main text, since $q_{0,0} = 0$, it follows that $Qq = \frac{1}{2}(\mathbb{I} - L_z)q$, such that (B.30) in Theorem B.5 directly reproduces (4.4) in Theorem 4.1.

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